

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 25

3-STAGE MYCORRHIZAL FUNGI PLANETARY ECOLOGICAL SEEDING PODS

Cyanobacteria, then fungi and moss, then grasses — iron feeds it



1

CYANOBACTERIA /
AI G&F — OXTD17F

2

FUNGI + MOSS —
SHATTER STONE TO

3

GRASSES / CONIFERS
— ANCHOR

IRON-RICH PLANETS — NATIVE RUST IS THE MICRONUTRIENT FUEL
POD ARCHITECTURE: PASS — OUTCOME IS SITE-DEPENDENT, TESTED PER WORLD

GP-IM-025

Revision v1.1 — 21 August 2026

26 of 290

VETTED: ARCHITECTURE PASS — OUTCOME SITE-TESTED

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LICENSE: CERN OPEN HARDWARE LICENCE v2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 25

PHASE 25: 3-STAGE MYCORRHIZAL FUNGI PLANETARY ECOLOGICAL SEEDING PODS — STATUS:

CURRENT. The terraformers' first wave: house-sized impact pods wrapped in Phase 13 ALON plating, dormant seeds encapsulated in an agar-cobalt radiation membrane, firing an automated three-stage terraforming sequence on iron-oxide-rich planets — cyanobacteria and algae oxidize the atmosphere; mycorrhizal mycelium and mosses shatter stone into fertile soil; vascular grasses and conifers anchor the new biosphere. And the ocean engine: phytoplankton dropped into newly melted basins on iron-rich worlds, native rust as the infinite micronutrient. Vetted: the pod ARCHITECTURE passes — the claimed ecological outcome is site-dependent and experimentally established per world, and that discipline is seated law of this chapter. [CORRECTED 21 AUGUST 2026 — 'house-sized' was Gemini's wording, never his: the pods are 100mm avocado-class, capsule-drone delivered, burn-zone targeted — §1.]

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-025, v1.1 — 21 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the seeding law as seated (contents line and both design bodies, verbatim) — §2 why it works (the biology, graded; the site-dependence discipline) — §3 the system on the ship — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Engineers specialized environmental seeding modules packed with highly resilient mycorrhizal fungi strains, designed to stabilize alien topsoil chemistry and initialize rapid terraforming baselines for pioneer outposts.' [VET CONSTRAINT, seated §5: 'rapid terraforming' is not a physics result — the engineering claim reads as 'deployable biological soil-conditioning system for tested compatible environments'; the line

stays preserved verbatim per doctrine.]

THE SEATED DESIGN BODY — THE SEED CORE (verbatim): 'Planetary Ecological Seed Core: House-sized impact pods wrapped in Phase 13 ALON plating containing dormant seeds encapsulated inside an agar-cobalt radiation membrane. Triggers an automated 3-stage terraforming explosion on iron-oxide rich planets: Stage 1 (Cyanobacteria/algae oxidize atmosphere); Stage 2 (Mycorrhizal fungi mycelium spores and mosses shatter stone into fertile soil); Stage 3 (Vascular grasses/conifers anchor the new biosphere).'

[AMENDED 21 AUGUST 2026 — 'House-sized' was Gemini's wording and he never noticed it; his pods were always avocado-size — 100mm, seated. Not impact warheads; no second ship to carry them. The §1 amendment governs the build.]

THE SEATED DESIGN BODY — THE IRON-PHYTOPLANKTON ENGINE (verbatim): 'Iron-Phytoplankton Engine: Identifies marine phytoplankton as the true global oxygen engine. Exoplanetary seeding pods drop phytoplankton directly into newly melted water basins on iron-rich planets (e.g., Mars), utilizing native rust as an infinite micronutrient fuel source to build a breathable atmosphere within generations.'

AMENDMENT — 21 AUGUST 2026 — THE 100MM LAW AND THE BURN-ZONE LAW (his ruling, verbatim): "25 hilarious "house sized" i never noticed gemini said that, i was imagining avocado size pods, not meteor sized impact warhead, not to mention another ship to carry them, additional spraying large regions with small packages will have more chance of striking a biocompatible surface medium, change that to 100mm, deployment is a capsule drone to get them through the atmosphere without burning up, then deploy them simply like backburn pods are dropped today, we are not seeding anything that has no burn zone to entre, because that is extremely low chance of creating an atmosphere, and the entire point"

SEATED AS THE 100MM LAW AND THE BURN-ZONE LAW (audit reading, disclosed): (1) THE SIZE, CORRECTED — 'house-sized' was Gemini's phrase and he never noticed it; his pods were always AVOCADO-SIZE — 100mm, seated. Not meteor-sized impact warheads, and no second ship needed to carry them. (2) THE SPRAY LOGIC — spraying large regions with small packages gives more chance of striking a biocompatible surface medium: numbers over mass, coverage over brute force. (3) THE DELIVERY CHAIN — a capsule drone carries the pods through the atmosphere without burning up, then deploys them simply, like backburn pods are dropped today. (4) THE BURN-ZONE LAW — we do NOT seed anything that has no burn zone to enter ('entre' preserved verbatim): an unburned target is an extremely low chance of creating an atmosphere, and atmosphere creation is the entire point. The fire comes first; the pods follow the burn. The design body's 'House-sized impact pods' stands as spoken — Gemini's words, preserved — with this amendment governing the build.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE BIOLOGY, AFFIRMED: mycorrhizal fungi genuinely interact with plant roots and soil chemistry — the underlying biology is real (the vet, seated). The three-stage sequence follows Earth's own ecological

succession: oxidizers first, soil-makers second, anchor plants third. The phytoplankton identification is also real: marine phytoplankton are Earth's true global oxygen engine, and iron fertilization of phytoplankton is established oceanography, not speculation.

THE SITE-DEPENDENCE DISCIPLINE, SEATED AS LAW OF READING: the problem word is 'alien'. No Earth organism can be ASSUMED to survive alien soil chemistry, pH, salinity, toxic metals, temperature, radiation, or the absence of compatible host plants (the vet's list, seated). So: the pod architecture passes; the ecological outcome is SITE-DEPENDENT and must be experimentally established per world. The claim reads: deployable biological soil-conditioning system for TESTED COMPATIBLE ENVIRONMENTS.

THE RADIATION MEMBRANE: dormant seeds ride inside an agar-cobalt membrane — the cobalt-bearing radiation discipline rhymes with the ship's own cobalt work (Phase 292's cobalt-silicone adhesive; the Phase 27 cone's cobalt-lead crystal) — cobalt compounds are the book's standing radiation-hard materials family.

THE HONEST TIMESCALE: 'within generations' is the seated horizon — atmosphere-building is generational engineering. the chapter certifies the architecture and the stage logic; the calendar belongs to the planet, not to the pod.

§3 — THE SYSTEM ON THE SHIP

- THE POD: house-sized impact capsule, Phase 13 AION plating wrapper, dormant seeds in the agar-cobalt membrane — built to arrive hard and alive. *[CORRECTED 21 AUGUST 2026 — 100mm avocado-class pods, capsule-drone delivered through the atmosphere, deployed like backburn drops; no hard arrival, no impact warhead.]*
- THE SEQUENCE: automated three-stage trigger on iron-oxide-rich ground — Stage 1 oxidizers (cyanobacteria/algae), Stage 2 soil-makers (mycorrhizal mycelium + mosses shattering stone), Stage 3 anchors (grasses/conifers).
- THE OCEAN HALF: seeding pods drop phytoplankton directly into newly melted water basins on iron-rich worlds — native rust is the micronutrient fuel; no imported fertilizer line.
- THE SELECTION LAW: per the site-dependence discipline — strains matched to the target world's TESTED compatibility before any pod flies (§9).
- THE BURN-ZONE LAW (his law, 21 August 2026): NEVER seed a target that has no burn zone to enter — an unburned target is an extremely low chance of creating an atmosphere, and atmosphere creation is the entire point. The fire comes first; the pods follow the burn.
- THE DELIVERY CHAIN (his law, 21 August 2026): a capsule drone carries the 100mm pods through the atmosphere without burning up, then deploys them simply — the backburn drop, scaled to a planet. Spray large regions with small packages: more chances to strike a biocompatible surface medium.

INTEGRATION NOTE (audit, flagged): the pods are impactors — launch, targeting and planetary-protection review ride the bench (§9); the ALON wrapper is the Phase 13 plating family (GP-IM-013 v1.4).

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE FEEDSTOCK CIRCLE : 'the harvested minerals seed self-sustaining biological food cultures — feeding the biological food-culture line (Phase 25 mycorrhizal fungi, Phase 26 sand-hydroponics) under ice, where sunlight never reaches' — the vent-minerals line feeds the fungi line, ship-side.
- THE CROSS-VET LAW : 'Cross-vets newly synthesized elements against Phase 25 mycorrhizal fungi, Phase 26 sand-hydroponics, and closed-loop cabin water lines' — new materials answer to the biology before they join the ship.
- THE CREW LINE : 'Biologists, horticulturalists, medics: 25/26 (fungi and sand hydroponics)...' — the phase owns crew posts, not just pods.
- THE BOOK'S OWN SUMMARY (seated): 'Phase 25 — Mycorrhizal Seeding Pods: the terraformers' first wave — fungi pods that make alien ground ready for Earth plants.' — read with §2's site-dependence discipline governing.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the pod: house-sized impact capsule in the Phase 13 ALON wrapper; impact survival is a structure problem — bench it (§9). *[CORRECTED 21 AUGUST 2026 — the pod is 100mm avocado-class; there is no impact survival problem — the capsule drone carries them through the atmosphere and drops them like backburn pods; the wrapper spec follows the 100mm form.]*
- STEP 2 — Build the membrane: the agar-cobalt encapsulation for dormant seed stock; dormancy, viability and radiation figures benched per strain (§9).
- STEP 3 — Stock the stages: Stage 1 oxidizers, Stage 2 fungi/mosses, Stage 3 anchors — strain libraries with compatibility records, not hopes.
- STEP 4 — Commission the sequence: the automated 3-stage trigger with environmental gates — each stage fires on measured conditions, not on a timer.
- STEP 5 — Write the site protocol: candidate world assay first (chemistry, pH, salinity, metals, temperature, radiation, water), strain match second, pod third — the site-dependence discipline made procedural.
- STEP 6 — Seed the ocean line: phytoplankton stock for iron-rich meltwater basins; the rust-

micronutrient budget verified per basin before release.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 13 — the 94mm stack (GP-IM-013 v1.4): the AION wrapper is the stack's plating family; impact duty rides its spec.
- PHASE 26 — the snowglobe (GP-IM-026): the cluster mate — the ship-side agriculture the seeding line feeds and learns from (§4).
- PHASE 12 — the agar bio-siphons (GP-IM-012): the agar discipline the membrane inherits; the bio-containment law the pods answer to.
- PHASE 292 — the cobalt-silicone adhesive: the cobalt radiation-materials family rhyme (§2).
- PHASE 27 — the deflector cone (GP-IM-027): cobalt-lead crystal — the same cobalt radiation family, shield-side.
- PHASE 59 — the ice-moon seeding: the crew doctrine pairs 25/26 with 59's ice work (§4).
- PHASE 24 — the scrub boundary (GP-IM-024 v1.0): every biological load crosses the boundary law, pods included.

§8 — DO-NOT-BUILD

- NEVER call it 'rapid terraforming' in spec text — the engineering claim is soil-conditioning for tested compatible environments (§2, §5).
- NEVER fly a pod to an unassayed world — the site protocol is seated law (§6, step 5).
- NEVER let a stage fire on a timer — environmental gates only; the planet sets the pace (§6, step 4).
- NEVER assume Earth biology survives alien ground — test, then match, then fly (§2).
- NO bypass of the boundary law — biological material crosses Phase 24 discipline in both directions (§7). The bench owes capsule-drone entry thermal figures, dispersal pattern density per region, 100mm pod membrane viability per strain, and the burn-zone assay (what counts as a burn zone to enter) alongside the candidate-world assay.
- NEVER call a pod house-sized — 100mm avocado-class, seated 21 August 2026 (§1).
- NEVER seed a target with no burn zone to enter — the burn zone is the entire point (§3).

§9 — OWED

THE BENCH, OWED: pod impact-survival figures (velocity, ground classes, AION wrapper thickness);

membrane dormancy/viability/radiation numbers per strain; strain library and compatibility records; stage-trigger environmental gate thresholds; the candidate-world assay spec (chemistry, pH, salinity, metals, temperature, radiation, water); phytoplankton basin budgets (iron availability, meltwater chemistry); planetary-protection review doctrine. All owed items ride the bench until spoken.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-025, v1.0 — CURRENT, seated law, master v2.1 (numerical print order, post-split). Built 20 August 2026. v1.1 (21 August 2026): the 100mm law, the delivery chain and the burn-zone law seated at §1 and §3; §9 re-cut to the drone-delivery benches.

PROVENANCE — THE STANDING ORDER, VERBATIM (seated in sitting 499): '16 and then bed for me, after we finish that you can do 17 to 27 and i will read all 10 first thing to see if we can get through them faster in blocks'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart. Next version triggers: the site protocol, the §9 benches, or his word.

END OF CHAPTER 25